

Q4指南

这个大题包括web crawler, fusion algorithm, query expansion, relevance feedback等概念考察，计算题考察排序算法（CombMNZ, Borda Fuse, CombSUM）

设计Web crawler遵守的标准 (24Q4.a)

1. Obeying **robots.txt** files and other methods to give instructions to crawlers.
2. **Not** overloading the website
3. **Not** consuming too much bandwidth.
4. Identifying who they are and who they belong to.

The Robots Exclusion Standard

- The Robots Exclusion Standard is a voluntary standard that asking web crawlers not to index certain portions of a web site.

Web crawler获取信息困难的情况 (4D) (24Q4.a)

1. Different URL: A Page can be represented in different URL.
2. Dynamic environment: Addition of new pages change the index of existing pages, need to revisited.
3. Defensive Design: Pages designed to reject automated access
4. Deep Web: Web crawlers cannot easily gain access to particular online resources.

Fusion algorithm利用的3种Effect (24Q4.c)

Effect	Definition	Application
Skimming Effect	Selecting top-ranked documents from each result set for fusion give better performance.	Used in all popular fusion algorithms .
Chorus Effect	When multiple retrieval systems agree on a document's relevance, it should be highly ranked in the fused results.	Important for Data Fusion .
Dark Horse Effect	Occurs when one system returns results of significantly different quality (either unusually accurate or inaccurate).	Very difficult to identify, generally not used

CombSUM and Borda Fuse用了三种Effect中的哪些? (18Q4.b)

CombSUM

把各系统中文档对应的分数相加得到最终分数

- **Skimming Effect**: 输入集中的高分会被转入结果集中
- **Chorus Effect**: 因为分数是累加计算的，多次出现的文档会得到更高的分数

Borda Fuse

- **Skimming Effect:** 排名越高的文档在 Borda 计分中获得的分数越高
- **Chorus Effect:** 如果一个文档在多个系统中排名靠前，其 Borda 总分将显著提升

延伸知识：为什么黑马效应没有被使用？如何使用黑马效应？

- 因为CombSUM和Borda Fuse默认所有系统权重一样，无法自动识别某个系统是否“异常可靠”或“异常不可靠”。
- 需要引入加权融合-Linear Combination

query expansion中synonyms的三种来源(24Q4.d)

1. Manual thesaurus (maintained by human editors)
2. Automatically derived thesaurus
3. Using word embeddings as a similar effect, without using a thesaurus

Rocchio Algorithm使用一个modified query vector提供RF，解释为什么Rocchio Algorithm可以提升IR系统有效性，这个modified query vector如何被计算 (23Q4.a)

$$\vec{q}_{opt} = \frac{1}{|C_r|} \sum_{\vec{d}_j \in C_r} \vec{d}_j - \frac{1}{|C_{nr}|} \sum_{\vec{d}_j \in C_{nr}} \vec{d}_j$$

- $\vec{q}$ is the original query provided by the user.
- D_r and D_{nr} are the documents that have been identified by the user as being relevant and nonrelevant respectively.
- α , β and γ are weights that can affect the behaviour of the formula.

Corpus Overlap (23Q4.c)

Different levels of corpus overlap can influence the design of a data fusion algorithm. Explain why this is the case.

类型	描述	文档重叠情况	Fusion方法
Disjoint Databases	各输入系统检索完全独立的文档集合，无共同文档。	无重叠（文档仅出现在一个系统中）	Collection Fusion
Identical Databases	各输入系统使用相同的文档集合，但采用不同检索算法。	完全重叠（文档可能出现在所有系统中）	Data Fusion
Overlapping Databases	各系统的文档集合部分重叠（非完全独立，也非完全一致）。	部分重叠（文档可能出现在部分系统中）	Difficult to draw reliable conclusions